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C2 — Elements, Compounds and Mixtures

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 1 & 3 (C2.1 Purity and separating mixtures; C2.2 Bonding; C2.3 Properties of materials)

Topic C2 explains what makes a substance **pure**, how mixtures are **separated**, how atoms are held together by **ionic, covalent and metallic bonding**, and how the type of bonding and structure decides a material's **properties** — including the carbon allotropes and nanoparticles. It is assessed on Paper 1 (Foundation) and Paper 3 (Higher). There is no Higher-tier-only content in C2; everything here can appear on both tiers.

C2.1 Purity and Separating Mixtures

What Does "Pure" Mean?

> In chemistry, a **pure substance** is a **single element or a single compound**, not mixed with any other substance. Elements and compounds are pure; **mixtures are impure**.

In everyday language "pure" means natural, clean or with nothing added (e.g. "pure orange juice", "pure mountain spring water"). Scientifically these are **mixtures**: orange juice contains water, sugars, acids and many other compounds; spring water contains dissolved minerals and gases. A solution is never pure — it is a mixture of solute and solvent.

Everyday use of "pure"	Scientific meaning
Nothing added; natural; unprocessed; "100% pure honey"	Contains only **one substance** (one element or one compound)
"Pure water" from a spring	Not pure — contains dissolved salts and gases. Only distilled water is (close to) pure H_2O.

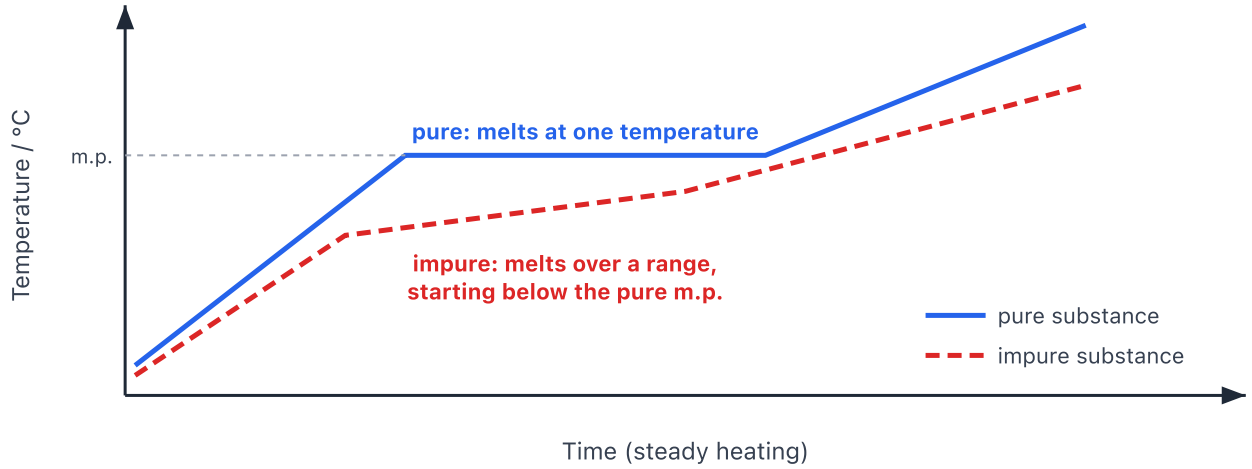
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Using Melting Point Data to Test Purity

Every pure substance has a **fixed, sharp melting point** (and boiling point) at a given pressure. Impurities change this:

- A **pure** substance melts at a **single, exact temperature** that matches the data-book value.
- An **impure** substance melts **over a range of temperatures**, and usually **below** the pure melting point (impurities lower the melting point and raise the boiling point).

Heating a Pure and an Impure Solid



Worked Example: Data book: pure aspirin melts at 136 °C. Sample A melts sharply at 136 °C; sample B melts between 128 °C and 133 °C. Which is pure?
Sample A is pure — it melts at a sharp, fixed temperature equal to the data value. Sample B is impure: it melts over a 5 °C range and below 136 °C.

Measuring melting point: a little powdered solid is packed into a thin glass capillary tube and heated slowly in an electric melting-point apparatus (or attached to a thermometer in an oil bath). The temperature is read when the solid first starts to melt and when it has all melted.

Relative Atomic, Molecular and Formula Mass

Relative atomic mass (A_r) — the weighted mean mass of an atom of an element, taking into account its naturally occurring isotopes, compared with 1/12 of the mass of an atom of carbon-12. (Given on the Periodic Table.)

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Relative molecular mass (M_r) — the sum of the relative atomic masses of all the atoms in a **molecule** (covalent substances, e.g. H_2O, CO_2).

Relative formula mass (M_r) — the sum of the relative atomic masses of all the atoms in the **formula** of a compound (used for ionic compounds and giant structures, e.g. NaCl, $CaCO_3$). Both have **no units**.

Worked Example: Chlorine is 75% chlorine-35 and 25% chlorine-37. Calculate its A_r.
$A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 = \mathbf{35.5}$.

Worked Example: Calculate the relative formula mass of calcium hydroxide, $Ca(OH)_2$. (A_r: Ca = 40, O = 16, H = 1)
The bracket doubles everything inside it: $Ca + 2 \times (O + H) = 40 + 2 \times (16 + 1) = 40 + 34 = \mathbf{74}$.

In a **balanced equation** the total M_r of the reactants equals the total M_r of the products (conservation of mass). Multiply each M_r by the big number in front of the formula:

Worked Example: Show that mass is conserved in $2Mg + O_2 \rightarrow 2MgO$. ($A_r$: Mg = 24, O = 16)
Reactants: $2 \times 24 + (2 \times 16) = 48 + 32 = 80$. Products: $2 \times (24 + 16) = 2 \times 40 = 80$. **80 = 80**, so mass is conserved.

Empirical Formulae from Atoms, Models and Diagrams

The **empirical formula** is the **simplest whole-number ratio** of atoms of each element in a compound. The **molecular formula** is the actual number of atoms of each element in one molecule.

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Model / molecule	Atoms counted	Empirical formula
Ethane ball and stick model (C_2H_6)	2 C, 6 H → 1 : 3	CH_3
Glucose, $C_6H_{12}O_6$	6 : 12 : 6 → 1 : 2 : 1	CH_2O
Hydrogen peroxide, H_2O_2	2 : 2 → 1 : 1	HO
Diagram of an ionic lattice with 6 X ions and 12 Y ions	6 : 12 → 1 : 2	XY_2
Carbon dioxide, CO_2	1 : 2 (already simplest)	CO_2

To find an empirical formula from a model: **count** the atoms of each element, write the ratio, then **divide by the highest common factor**. Going the other way, if you know the empirical formula and the M_r, the molecular formula is a whole-number multiple: e.g. empirical formula CH_2 with $M_r = 56$ → ($CH_2 = 14$) → 56 ÷ 14 = 4 → C_4H_8. For giant structures (ionic lattices, diamond, SiO_2) only the empirical formula can be given, because there are no separate molecules.

Formulations

> A **formulation** is a **mixture designed as a useful product**. Each component is added in a carefully measured quantity for a particular purpose.

In an **alloy** (e.g. steel = iron + carbon; brass = copper + zinc), atoms of different sizes distort the regular layers of the metal, so the layers cannot slide over each other as easily — the alloy is **harder and stronger** than the pure metal. Other examples: paints, medicines (tablets with the active drug plus fillers, binders and coatings), fertilisers, cleaning products, cosmetics, fuels and foods. Many useful materials are formulations rather than pure substances, because the mixture has better properties than any single component.

Separation Techniques

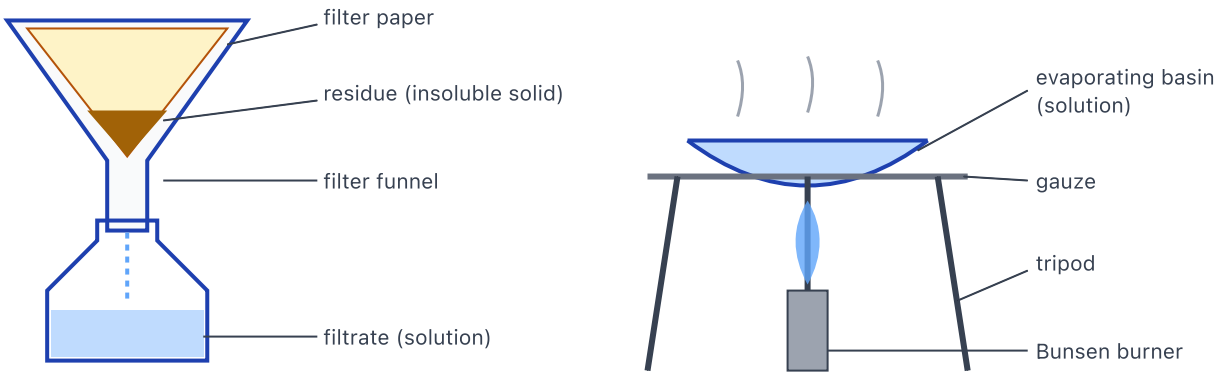
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Mixtures can be separated by **physical** methods because the substances in them are not chemically bonded. The method depends on the property that differs between the components.

Filtration and Crystallisation

- **Filtration** separates an **insoluble solid** from a liquid (e.g. sand from salty water). The liquid passes through the tiny holes in the filter paper (the **filtrate**); the solid particles are too large and stay behind (the **residue**).
- **Crystallisation** obtains a **soluble solid** from a solution (e.g. copper sulfate crystals). Heat the solution in an evaporating basin to evaporate some of the solvent until the solution is **saturated** (crystals start to form on a glass rod dipped in and removed). Leave it to cool slowly; crystals form as the solubility falls. Filter off the crystals and dry them between filter papers or in a warm oven.
- **Evaporation to dryness** is quicker but gives small crystals and can decompose some salts (e.g. hydrated salts lose their water of crystallisation) — crystallisation gives larger, purer crystals.

Filtration and Crystallisation



1. Filtration

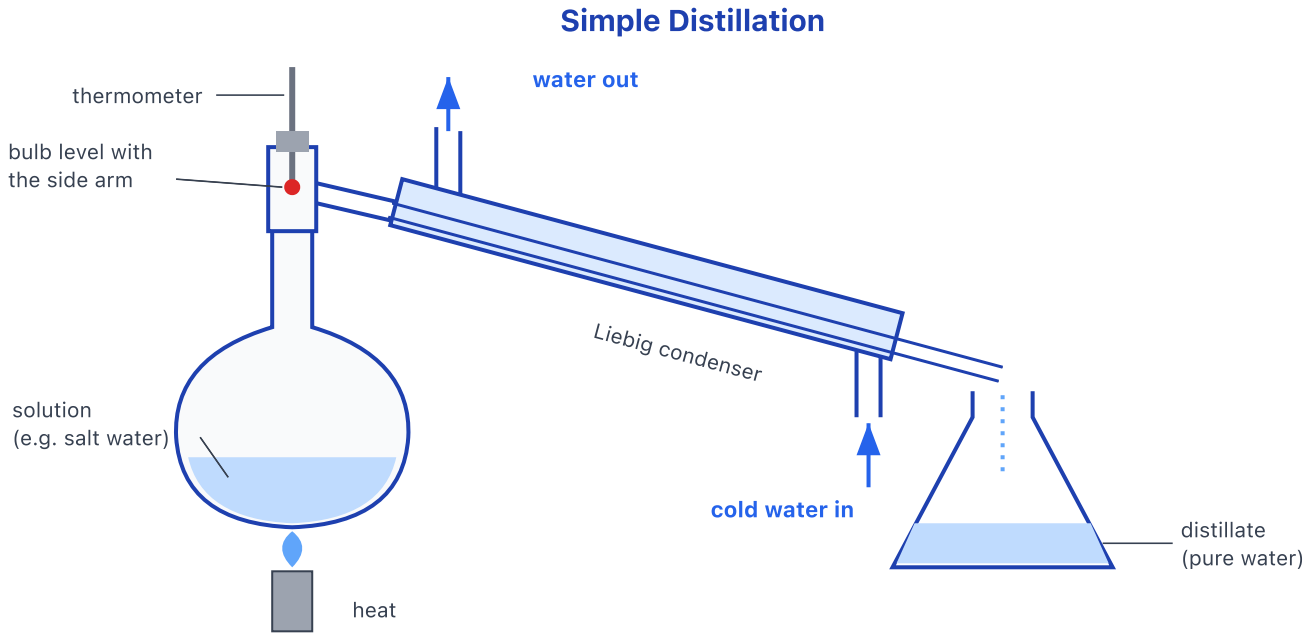
2. Crystallisation

heat until saturated → cool → filter → dry crystals
(a water bath gives gentler heating)

Simple Distillation

Simple distillation separates a **liquid (solvent) from a solution**, e.g. pure water from salty water or ink. The solution is heated; the solvent boils and its vapour passes into the **condenser**, where it is cooled and condenses back to a liquid (the **distillate**). The dissolved solid stays in the flask.

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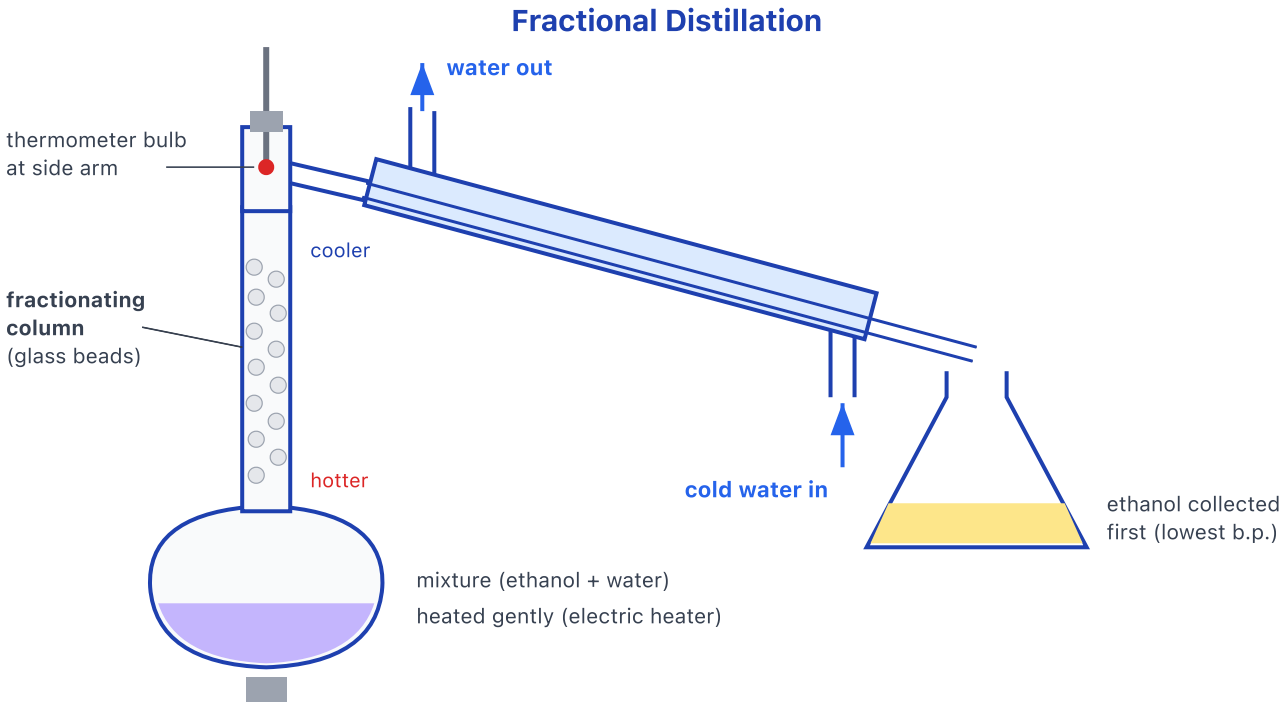
Exam Tip: Two classic diagram marks: the **thermometer bulb is level with the side arm** (it measures the temperature of the vapour going into the condenser, not of the liquid), and **cold water enters the condenser at the bottom** (lower end) and leaves at the top, so the jacket stays completely full of cold water. The collecting flask is not sealed, and the condenser outlet does not dip into the distillate.

Fractional Distillation

Fractional distillation separates a **mixture of miscible liquids with different boiling points** (e.g. ethanol, b.p. 78 °C, and water, b.p. 100 °C; or crude oil into fractions).

1. The mixture is heated; vapours of both liquids rise up the **fractionating column** (packed with glass beads, which give a large surface area).
2. The column is **hotter at the bottom and cooler at the top**. Vapour of the higher-boiling liquid (water) condenses on the beads and drips back into the flask.
3. The liquid with the **lowest boiling point** reaches the top first, passes into the condenser and is collected. The thermometer stays at its boiling point (78 °C) while it distils.
4. When the temperature rises, change the collecting flask — the next liquid is now distilling.

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PAG C4 — Distillation: distil a mixture (e.g. orange juice, cherry cola, inks or a hydrocarbon mixture) using simple or fractional distillation. Use a Bunsen burner, water bath or electric heater (flammable liquids such as ethanol must *not* be heated with a naked flame), record the temperature at which each fraction distils, and test the distillate (e.g. boiling point or density) to check its purity. Anti-bumping granules give smooth boiling.

Chromatography

Chromatography separates mixtures of soluble substances (e.g. dyes in ink, food colourings, amino acids) and is used to **identify** substances and test their **purity**.

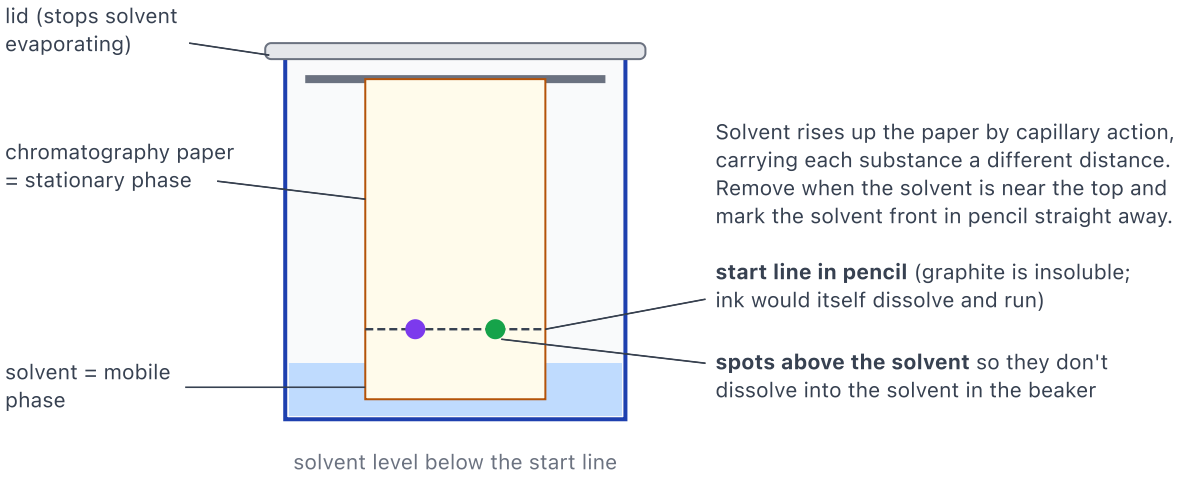
All chromatography has two phases:

- a **stationary phase** that does not move — the paper (paper chromatography) or the thin layer of silica / alumina on a glass, metal or plastic plate (TLC).
- a **mobile phase** that moves through it — the **solvent** (water, ethanol or another solvent) in paper chromatography and TLC, or an unreactive carrier gas in gas chromatography.

Separation depends on how each substance is **distributed between the two phases**. A substance that is more attracted to (spends more time in) the mobile phase travels further; one that is more strongly attracted to the stationary phase travels a shorter distance.

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Setting Up Paper Chromatography



Paper vs Thin Layer Chromatography (TLC)

	Paper chromatography	Thin layer chromatography (TLC)
Stationary phase	Chromatography paper (cellulose fibres holding water)	Thin layer of silica gel or alumina on a glass, plastic or aluminium plate
Mobile phase	Aqueous (water) or non-aqueous solvent (e.g. ethanol, propanone)	Usually a non-aqueous (organic) solvent
Advantages	Cheap and simple	Faster, better separation, sharper spots, smaller samples; wider choice of solvents

Choice of solvent: water (aqueous) is used for water-soluble substances such as food colourings and some inks. Substances that do not dissolve in water (e.g. permanent-marker ink, chlorophyll, many organic compounds) need a **non-aqueous solvent**, such as ethanol or propanone — these are flammable, so no naked flames.

Locating agents: colourless substances (e.g. amino acids, sugars) do not show up as spots. After running, the chromatogram is sprayed with a **locating agent** that reacts to form coloured spots (e.g. **ninhydrin** turns amino acids purple), or a TLC plate containing a fluorescent dye is viewed under **ultraviolet light**, or placed in a tank of iodine vapour.

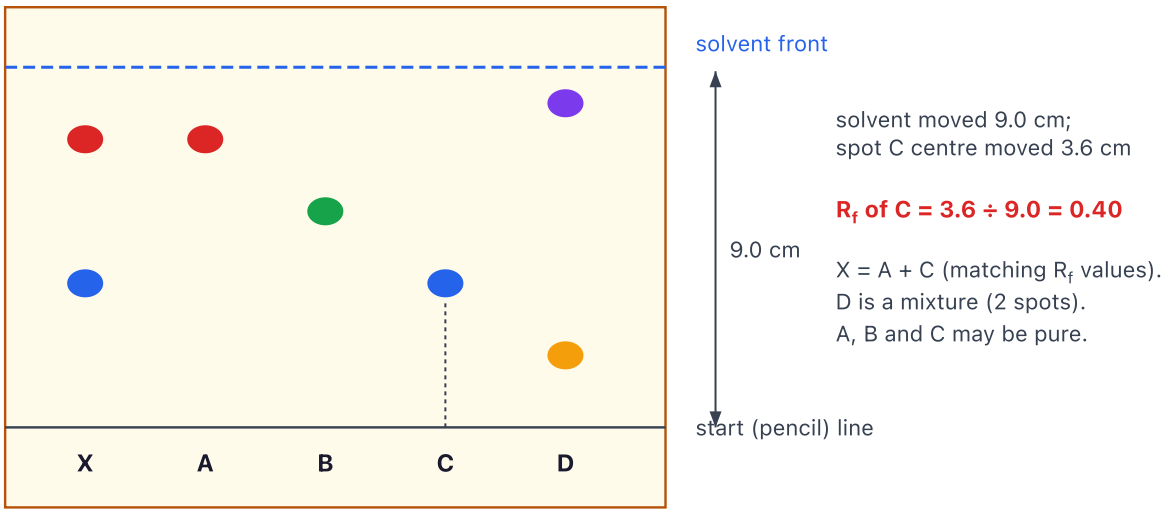
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Interpreting Chromatograms and R_f Values

R_f = distance moved by substance ÷ distance moved by solvent

(both measured from the start line; R_f has no units and is always less than 1)

A Chromatogram: Which Dyes Are in Ink X?



- A **pure** substance gives **one spot** (in every solvent tried). A **mixture** gives **two or more** spots.
- Two spots that have travelled the **same distance** (same R_f) in the same solvent are probably the **same substance**. Compare with known reference substances run on the same paper.
- R_f values depend on the solvent, the stationary phase and the temperature, so compare values obtained under the **same conditions**.
- A spot that stays on the start line is **insoluble** in that solvent.

Worked Example: In a chromatogram the solvent front is 8.0 cm from the start line. A spot has an R_f of 0.65. How far did it travel?

Rearrange: distance moved by substance = R_f × distance moved by solvent = 0.65 × 8.0 = **5.2 cm**.

Gas Chromatography (GC)

In **gas chromatography** the mobile phase is an unreactive **carrier gas** (e.g. nitrogen or helium) and the stationary phase is a liquid or solid coating inside a long, thin, coiled column in an oven. The vaporised sample is carried through; each substance takes a different time to come out — its **retention time**. A detector records a peak for each substance. The number of peaks shows how many substances are present, the retention time helps identify them (compared with standards), and the peak size shows relative amounts. GC is very sensitive and works with tiny samples — useful for testing purity of fuels, drugs, food flavourings and for detecting drugs in blood.

Choosing Chromatography to Test Purity

Which method? Paper chromatography suits coloured, soluble substances (pure = a single spot in several different solvents); TLC suits small samples and colourless organic compounds (with a locating agent or UV); GC suits volatile substances and tiny amounts (pure = a single peak at the retention time of the pure substance).

> **PAG C3 — Separation techniques (chromatography):** use paper or thin layer chromatography to identify the mixture of dyes in an unknown ink. Draw a pencil start line ~1.5 cm from the bottom, spot the unknown and reference dyes, stand the paper in a shallow layer of solvent (below the line) in a covered beaker, then mark the solvent front, measure distances and calculate R_f values.

Choosing a Purification Technique

Mixture	Property that differs	Technique
Insoluble solid + liquid (sand + water; a precipitate)	Solubility / particle size	**Filtration**
Soluble solid from a solution (salt from salt water)	Solid doesn't evaporate	**Crystallisation** (or evaporation)
Solvent from a solution (pure water from sea water)	Boiling point (solvent much lower)	**Simple distillation**

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Miscible liquids (ethanol + water; crude oil)	Boiling points (fairly close)	**Fractional distillation**
Several dissolved substances (dyes in ink)	Solubility / attraction to phases	**Chromatography**
Two solids, one soluble (salt + sand)	Solubility	Dissolve in water → **filter** → **crystallise** the filtrate

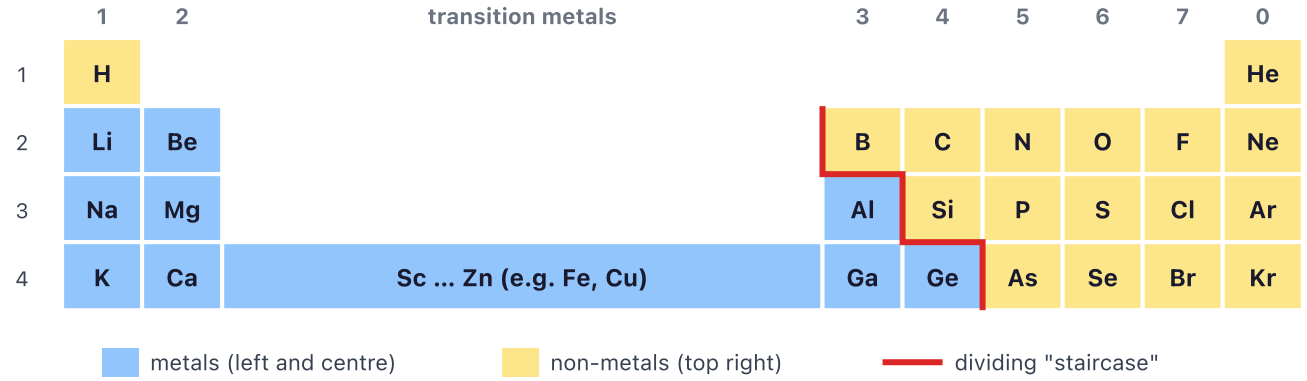
Worked Example: Suggest how to obtain pure, dry sodium chloride from rock salt (salt mixed with sand and grit).
1. Crush the rock salt and stir with warm water: the salt **dissolves**, sand does not. 2. **Filter**: sand stays as residue; salt solution is the filtrate. 3. Heat the filtrate in an evaporating basin until saturated, then leave to **crystallise**. 4. Filter off the crystals and **dry** them. Check purity by melting point (801 °C).

PAG C7 — Production of salts (also linked to C2.1 purification): make a pure, dry sample of a soluble salt, e.g. copper sulfate from excess copper oxide and warm dilute sulfuric acid — filter off the excess solid, then crystallise the filtrate. The full method is covered in Topic C3.

C2.2 Bonding

Metals and Non-Metals

Metals and Non-Metals in the Periodic Table



Periods 1–4 shown (period numbers on the left). Elements next to the staircase (e.g. Si, Ge) have in-between properties.

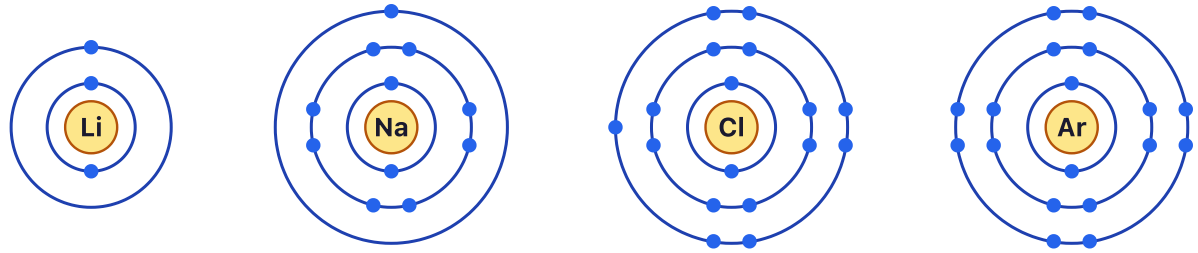
Property	Metals	Non-metals
Appearance	Shiny (lustrous) when freshly cut	Usually dull
State at room temperature	Solid (except mercury)	Many are gases; some solids; bromine is liquid
Melting / boiling point	Usually high	Usually low (except giant covalent C, Si)
Density	Usually high	Usually low
Conductivity	Good conductors of heat and electricity	Poor conductors / insulators (except graphite)
Mechanical	Malleable and ductile; sonorous	Brittle when solid
Ions formed	**Positive ions** (lose electrons)	**Negative ions** (gain electrons), or share electrons
Reaction with oxygen	Form **basic** oxides (e.g. MgO, Na_2O); soluble ones give alkaline solutions	Form **acidic** oxides (e.g. CO_2, SO_2, P_4O_{10}) that give acidic solutions
Typical reaction	$4Na + O_2 \rightarrow 2Na_2O$; $Mg + 2HCl \rightarrow MgCl_2 + H_2$	$C + O_2 \rightarrow CO_2$; $S + O_2 \rightarrow SO_2$

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Atomic Structure and the Periodic Table

Electrons occupy **shells (energy levels)**. For the first 20 elements: the first shell holds up to **2** electrons, the second up to **8**, and the third up to **8** (then the fourth starts). The innermost shells fill first.

Electron Arrangements and Position in the Periodic Table



Li (3) 2,1
Group 1 · Period 2

Na (11) 2,8,1
Group 1 · Period 3

Cl (17) 2,8,7
Group 7 · Period 3

Ar (18) 2,8,8
Group 0 · Period 3

Number of shells = period number · Electrons in outer shell = group number (Group 0: full outer shell)

- Elements are arranged in order of **atomic number** (number of protons = number of electrons in the atom).
- The **period** (row) number = the **number of occupied shells**. Sodium (2,8,1) has 3 shells → Period 3.
- The **group** number = the **number of electrons in the outer shell** (for Groups 1–7). Chlorine (2,8,7) → Group 7. Group 0 atoms have full outer shells (helium has 2).
- **Metals** are on the left: their atoms have **few outer electrons** (1, 2 or 3) that are lost to form positive ions. **Non-metals** are on the right: their atoms have **many outer electrons** (4–7), and they gain or share electrons.

Electron arrangements, Z = 1–20: H 1 · He 2 · Li 2,1 · Be 2,2 · B 2,3 · C 2,4 · N 2,5 · O 2,6 · F 2,7 · Ne 2,8 · Na 2,8,1 · Mg 2,8,2 · Al 2,8,3 · Si 2,8,4 · P 2,8,5 · S 2,8,6 · Cl 2,8,7 · Ar 2,8,8 · K 2,8,8,1 · Ca 2,8,8,2. (Hydrogen, 1, is a non-metal usually placed on its own.)

Reactions and Electron Arrangement

The **chemical reactions of an element depend on its outer-shell electrons**, which depend on its atomic number. Atoms react to gain a stable, full outer shell (like the noble gases):

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- **Group 0** atoms already have full outer shells, so they are **unreactive** (inert) and exist as single atoms.
- **Group 1** atoms have 1 outer electron, easily **lost** → 1+ ions. All Group 1 elements react in similar ways (e.g. with water to form hydroxides + hydrogen) because they have the same number of outer electrons.
- **Group 2** atoms lose 2 electrons → 2+ ions. **Group 3** (Al) → 3+.
- **Group 6** atoms gain 2 electrons → 2– ions; **Group 7** atoms gain 1 electron → 1– ions (or share electrons with other non-metals).
- Elements in the **same group have similar chemical properties** because they have the same number of outer electrons.

Three Types of Strong Bond

Bond	Between	What happens to electrons	Electrostatic attraction
Ionic	Metal + non-metal	**Transferred** from metal to non-metal atoms	Between oppositely charged ions, in all directions
Covalent	Non-metal + non-metal	**Shared** in pairs	Between the shared pair of electrons and the two positive nuclei
Metallic	Metal atoms	Outer electrons **delocalised** (free to move through the structure)	Between positive metal ions and the "sea" of delocalised electrons

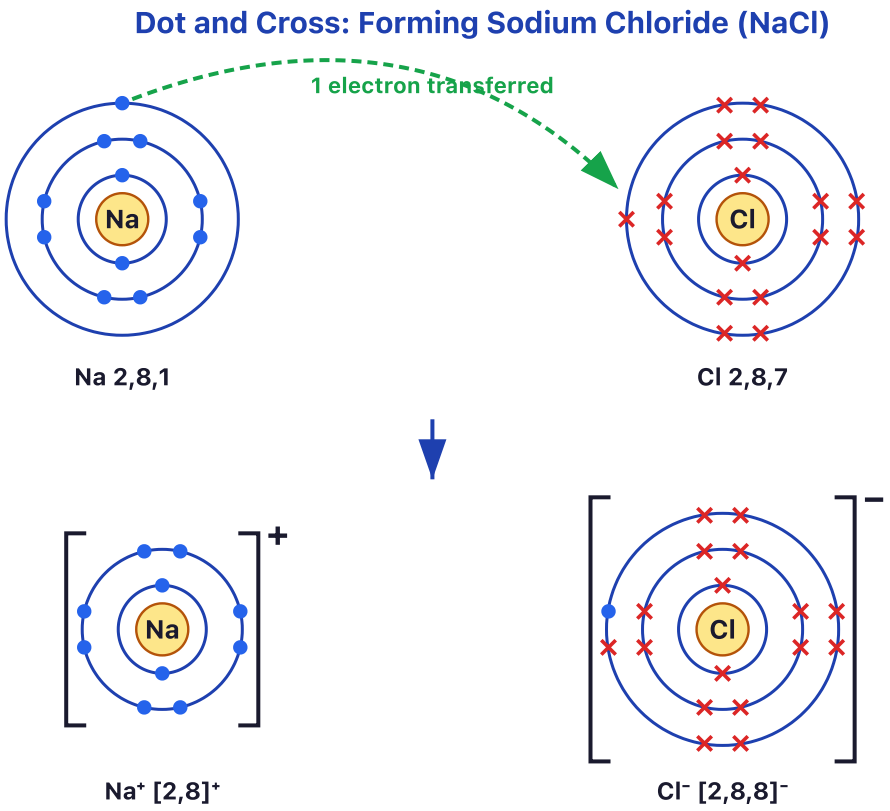
> **Exam Tip:** All chemical bonds are **electrostatic forces** of attraction between opposite charges — a bond is not a physical "stick". For covalent bonds, say the attraction is between the **shared pair of electrons and the nuclei** of both atoms.

Ionic Bonding

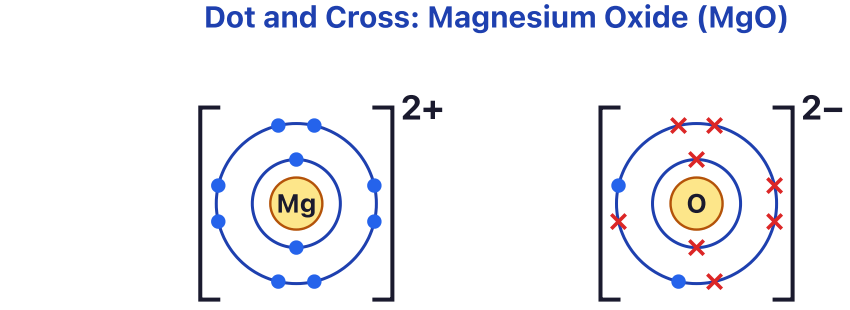
When a metal reacts with a non-metal, electrons are **transferred** from the metal atoms to the non-metal atoms. Both form ions with full outer shells (the electron arrangement of a noble gas). The

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oppositely charged ions are held together by strong **electrostatic forces of attraction** — the **ionic bond**.



● = sodium electron × = chlorine electron. Both ions now have full outer shells.



Magnesium oxide, MgO: Mg (2,8,2) gives 2 electrons to O (2,6)

● = magnesium electron × = oxygen electron. 2 electrons transferred; charges 2+ and 2− balance.

When drawing dot and cross diagrams for **binary ionic compounds** (two elements):

1. Work out the ion charges from the groups (Group 1: +, Group 2: 2+, Group 6: 2−, Group 7: −).
2. Balance the charges to get the formula ($MgCl_2$ needs two Cl^- for one Mg^{2+}).
3. Draw each ion with its new electron arrangement; use dots for one atom's electrons and crosses for the other's. The metal ion usually has an empty outer shell, so draw its full inner

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shell (or show it as empty).

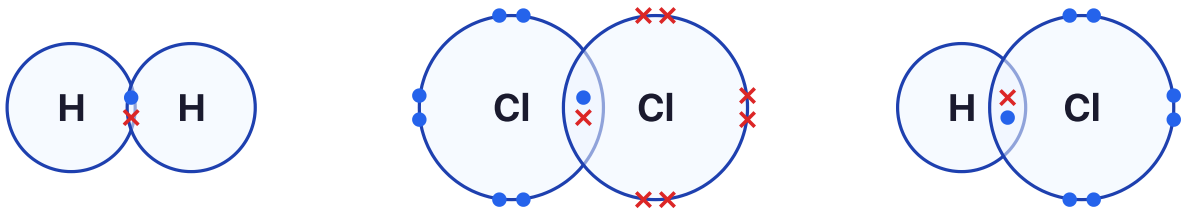
4. Put each ion in square brackets with its charge at the top right.

Compound	Ions	Formula	Electrons transferred
Sodium chloride	Na^+, Cl^-	NaCl	1 (Na → Cl)
Magnesium oxide	Mg^{2+}, O^{2-}	MgO	2 (Mg → O)
Calcium chloride	Ca^{2+}, Cl^-	$CaCl_2$	1 to each of 2 Cl
Sodium oxide	Na^+, O^{2-}	Na_2O	1 from each of 2 Na
Lithium fluoride	Li^+, F^-	LiF	1 (Li → F)

Covalent Bonding

When non-metal atoms bond, they **share pairs of electrons** so each atom gets a full outer shell. A shared pair is a **single covalent bond**; two shared pairs make a **double bond**; three make a **triple bond**. Covalent bonds are **strong**. Pairs of outer electrons not involved in bonding are called **lone pairs**.

Dot and Cross: H_2, Cl_2 and HCl (single bonds)



hydrogen, H_2
H–H

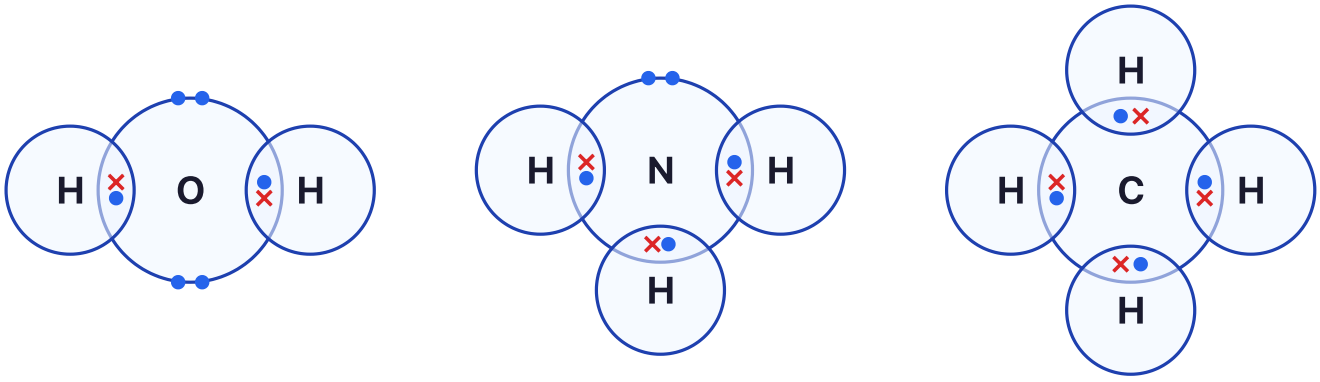
chlorine, Cl_2
Cl–Cl

hydrogen chloride, HCl
H–Cl

Outer shells only. ● and × show electrons from different atoms; each overlap holds the shared pair(s).

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Dot and Cross: H_2O, NH_3 and CH_4



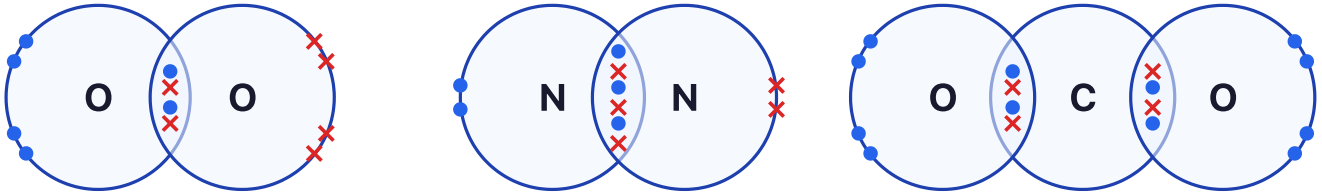
water, H_2O
2 bonding pairs, 2 lone pairs on O

ammonia, NH_3
3 bonding pairs, 1 lone pair on N

methane, CH_4
4 bonding pairs

Outer shells only. ● and × show electrons from different atoms; each overlap holds the shared pair(s).

Dot and Cross: Multiple Bonds in O_2, N_2 and CO_2



oxygen, O_2
O=O double bond

nitrogen, N_2
N≡N triple bond

carbon dioxide, CO_2
O=C=O

Outer shells only. ● and × show electrons from different atoms; each overlap holds the shared pair(s).

Molecule	Bonds	Lone pairs on central atom	Check: outer electrons around each atom
H_2, Cl_2, HCl	1 single bond	Cl: 3 lone pairs	H: 2 · Cl: 8
H_2O	2 single	2	O: 8 (2 bonds + 2 lone pairs)
NH_3	3 single	1	N: 8 (3 bonds + 1 lone pair)
CH_4	4 single	0	C: 8 (4 bonds)
O_2	1 double	2 on each O	O: 8
N_2	1 triple	1 on each N	N: 8

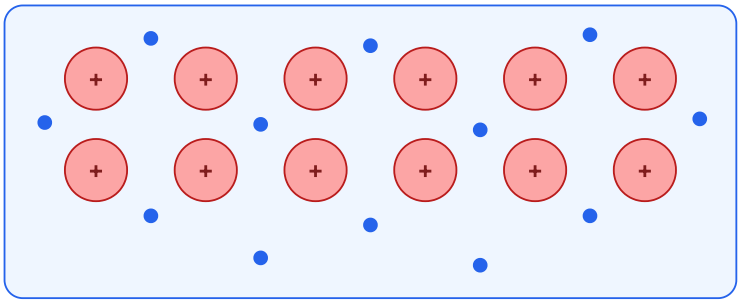
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CO_2	2 double	C: 0 · each O: 2	C: 8 · O: 8

Exam Tip: Count before you finish a dot and cross diagram: every H must have 2 electrons, every other atom 8 in its outer shell, and the total number of dots must equal the outer electrons of that element (O = 6, N = 5, C = 4, Cl = 7). Shared pairs go in the overlap of the circles.

Metallic Bonding

Metallic Bonding: Positive Ions in a Sea of Delocalised Electrons

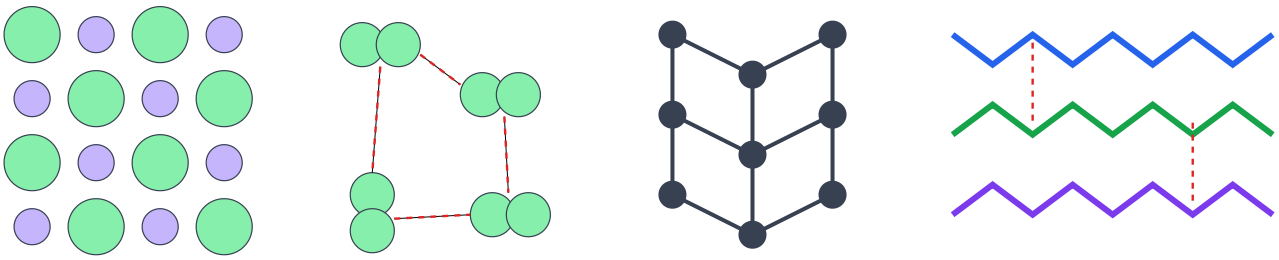


Regular layers of positive metal ions; outer electrons (blue) delocalised and free to move.

In a metal, the atoms are packed in a regular **giant lattice**. Each atom's outer-shell electrons become **delocalised** — free to move through the whole structure. The **metallic bond** is the strong electrostatic attraction between the **positive metal ions** and the **sea of delocalised electrons**.

Comparing the Five Types of Structure

Giant Ionic, Simple Molecular, Giant Covalent and Polymer Structures



Giant ionic (NaCl)
green Cl^-, purple Na^+
regular 3D lattice

Simple molecules (Cl_2)
strong covalent bonds inside;
weak forces between molecules

Giant covalent
all atoms joined by
covalent bonds (SiO_2)

Polymer (poly(ethene))
long chains of covalently bonded
atoms; forces between chains

Red dashed lines = intermolecular forces (weak). Solid lines = covalent bonds (strong).

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Structure	Particles	Bonds and their arrangement	Examples
Ionic compound	Positive and negative ions	Giant 3D lattice; strong electrostatic attraction between oppositely charged ions in **all directions**	NaCl, MgO, $CaCl_2$
Simple molecules	Small molecules	Strong covalent bonds **within** each molecule; **weak intermolecular forces between** molecules	H_2O, CO_2, CH_4, Cl_2, O_2
Giant covalent	Atoms	Huge network: **every atom joined to others by strong covalent bonds**; no separate molecules	Diamond, graphite, silicon dioxide (SiO_2)
Polymers	Very long molecules (chains)	Strong covalent bonds **along each chain**; intermolecular forces between chains (stronger than in small molecules because chains are large)	Poly(ethene), PVC, nylon
Metals	Positive ions + delocalised electrons	Giant lattice; strong electrostatic attraction between positive ions and the sea of delocalised electrons	Copper, iron, aluminium

Limitations of Representations and Models

Every way of drawing a structure shows some features and hides others:

Representation	Shows	Limitations
Dot and cross diagram	Which atom each electron came from; how electrons are shared / transferred; the formula	2D only — does not show the **shape** of the molecule or the 3D lattice; electrons from different atoms are actually identical; does not show relative sizes of atoms / ions; for ionic compounds shows only one formula unit, not the giant lattice

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Ball and stick model	3D arrangement and shape; bond angles; how atoms are connected	Atoms are not to scale and are shown too far apart (there are no big gaps really); bonds are not physical sticks; doesn't show which electrons are involved or the difference between single and double bonds clearly; for ionic lattices, "sticks" wrongly suggest bonds only between neighbours in fixed directions
2D displayed formula (e.g. H–O–H)	Which atoms are bonded; single / double bonds	No 3D shape or bond angles (water is bent, not straight); no lone pairs
Molecular formula (C_2H_6)	Number of each type of atom	No information on how atoms are arranged

> **Practical suggestion (C2.2):** build ball and stick models of molecules such as CH_4, H_2O, NH_3, CO_2 and C_2H_4 with molecular model kits, and compare them with the dot and cross diagrams to see what each representation shows and hides.

Mendeleev and the Modern Periodic Table

In 1869 Dmitri Mendeleev arranged the 60 or so known elements in order of **increasing atomic weight** (relative atomic mass), and put elements with **similar properties** in the same vertical group. His key ideas:

- He **left gaps** for elements not yet discovered, and **predicted their properties** from the elements around them. When "eka-silicon" (germanium) was discovered in 1886, its properties closely matched his predictions — strong evidence for his table.
- He sometimes **swapped the order** of pairs of elements so that they fell in the group with similar properties, even though this broke the atomic-weight order: e.g. tellurium (A_r 127.6) was placed *before* iodine (A_r 126.9) so that iodine sits with the other halogens.

In the early 20th century, the discovery of **protons** and the measurement of the **atomic number** of each element (Moseley, 1913) showed that the true order is by **atomic number**, not atomic weight. Ordering by atomic number:

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- puts every element in the group with similar properties **without any swaps** (Te/I, Ar/K, Co/Ni all fall into place);
- explains the anomalies: atomic weight depends on **isotopes** (numbers of neutrons), which Mendeleev could not know about;
- links position to **electron arrangement**: the group = outer electrons, so elements in a group react similarly;
- confirms there are no further missing elements between known ones. The noble gases (Group 0), discovered later, slotted in as a complete new group.

C2.3 Properties of Materials

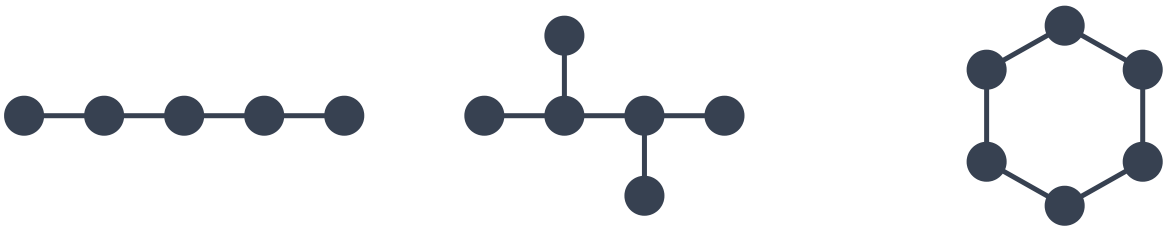
Carbon: Chains, Rings and Families of Compounds

A carbon atom has 4 outer electrons (2,4), so it **forms four covalent bonds**. Carbon atoms can bond strongly to each other as well as to other elements (H, O, N, Cl...). This means carbon can form:

- **long chains** (straight or branched) of almost any length;
- **rings** of carbon atoms;
- single, double and triple bonds;
- **families** (homologous series) of similar compounds, such as the alkanes and alkenes, with the same general formula and similar chemical properties.

This explains the **vast array of natural and synthetic organic compounds** — millions of them, including fuels, plastics, medicines, proteins, fats, sugars and DNA.

Carbon Skeletons: Chains, Branches and Rings



straight chain
pentane, C_5H_{12}

branched chain
C_6H_{14} isomer

ring
cyclohexane, C_6H_{12}

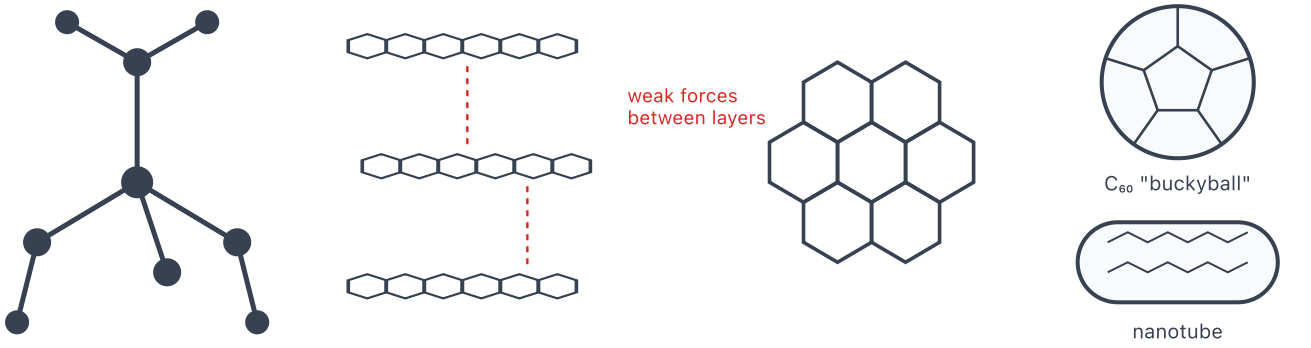
Carbon atoms only; each C also bonds to enough H atoms to make 4 bonds in total.

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Allotropes of Carbon

Allotropes are different structural forms of the same element. Carbon's allotropes have very different properties because their atoms are bonded and arranged differently.

Diamond, Graphite, Graphene and Fullerenes



Diamond
each C bonded to 4 others
giant 3D network

Graphite
each C bonded to 3 others;
layers; delocalised electrons

Graphene
a single layer of graphite,
one atom thick

Fullerenes
hollow cages / tubes of C;
rings of 5, 6 or 7 atoms

Solid lines = strong covalent bonds. Red dashed = weak forces between graphite layers.

Allotrope	Structure and bonding	Properties	Uses
Diamond	Giant covalent; each C forms **4 strong covalent bonds** in a rigid 3D network; no free electrons	Very **hard**; very high melting point (lots of strong bonds to break); does **not conduct** electricity (no delocalised electrons); insoluble	Cutting tools, drill tips, jewellery
Graphite	Giant covalent; each C forms **3 covalent bonds** in flat hexagonal layers; **weak forces between layers**; one **delocalised electron** per carbon atom	Very high melting point; **soft and slippery** (layers slide over each other); **conducts electricity** (delocalised electrons move along the layers)	Electrodes, lubricant, pencil "lead"
Graphene	A **single layer** of graphite — one atom thick; 3 bonds per	Extremely strong for its mass; very light;	Electronics, composites,

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	C; delocalised electrons	excellent electrical and thermal conductor; transparent and flexible	touchscreens (developing)
Fullerenes	**Simple molecules** (e.g. C_{60}, Buckminsterfullerene) or long **nanotubes**; hollow cages of hexagonal (and pentagonal) rings; delocalised electrons	C_{60}: weak intermolecular forces → low melting point, slippery. Nanotubes: very high tensile strength, conduct electricity, very high length : diameter ratio	Drug delivery (cage), lubricants, catalysts, reinforcing materials (e.g. tennis rackets), nanoelectronics

Exam Tip: Graphite conducts because it has **delocalised electrons** that can move through the structure — not "free ions". It is soft because the **forces between layers are weak**, not because covalent bonds are weak. Its melting point is high because melting means breaking the strong covalent bonds within the layers.

Bonds, Intermolecular Forces and Changes of State

To melt or boil a substance, energy must be transferred to overcome the forces holding the particles together. **The stronger the forces that must be overcome, the more energy is needed and the higher the melting / boiling point.**

Simple molecular substances (H_2O, CO_2, Cl_2) have **low** melting and boiling points: only the **weak intermolecular forces** between molecules are overcome — the strong covalent bonds inside the molecules are **not** broken. **Polymers** have stronger forces between their long chains, so they are solids. **Ionic** compounds (NaCl, m.p. 801 °C), **metals** (iron, 1538 °C) and **giant covalent** structures (diamond, > 3500 °C) have **high** melting points because many strong ionic, metallic or covalent bonds must be broken, which needs a lot of energy.

For simple molecules, intermolecular forces **increase with the size of the molecule**, so larger molecules have higher boiling points: CH_4 (–162 °C) < C_8H_{18} (126 °C); F_2 and Cl_2 are gases, Br_2 is a liquid and I_2 is a solid at room temperature.

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Exam Tip: The #1 bonding mistake: "water boils at 100 °C because the covalent bonds break". Boiling water gives steam — still H_2O molecules. Only the **intermolecular forces between the molecules** are overcome. Breaking the O–H bonds would make new substances (H and O).

Predicting States from Data

Compare the given temperature with the melting point (m.p.) and boiling point (b.p.):

temperature < m.p. → **solid** · m.p. < temperature < b.p. → **liquid** · temperature > b.p. → **gas**

Worked Example: Use the data to give the state of each substance at room temperature (20 °C) and at 100 °C.

Substance	m.p. / °C	b.p. / °C	At 20 °C	At 100 °C
Oxygen	−218	−183	gas	gas
Bromine	−7	59	liquid	gas
Ethanol	−114	78	liquid	gas
Phenol	41	182	solid	liquid
Sodium chloride	801	1413	solid	solid

Watch negative numbers carefully: −7 °C is *below* 20 °C, so bromine has already melted at room temperature.

Bulk Properties and Structure

The properties of a **bulk material** come from the way huge numbers of particles are bonded and arranged — **individual atoms do not have bulk properties**. A single copper atom is not "shiny", "ductile" or "a conductor", and a single carbon atom is not "hard" — these are properties of the

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whole structure (diamond and graphite are both made only of carbon atoms but are completely different).

Structure	Melting / boiling point	Conducts electricity?	Other properties
Ionic	High — many strong electrostatic forces in all directions	**Not when solid** (ions fixed in the lattice); **yes when molten or dissolved** (ions free to move and carry charge)	Hard but brittle (layers shift so like charges line up and repel); many dissolve in water
Simple molecular	Low — weak intermolecular forces	No — molecules have no overall charge; no free electrons or ions	Often gases or liquids; solids are soft
Giant covalent	Very high — many strong covalent bonds	No (except graphite and graphene — delocalised electrons)	Hard (diamond, SiO_2); insoluble
Polymers	Usually solids; soften / melt over a range of temperatures	No (usually insulators)	Flexible or rigid depending on chain length and cross-links; low density
Metals	Usually high — strong metallic bonding	**Yes, solid and liquid** — delocalised electrons move through the structure	Good thermal conductors (delocalised electrons); **malleable and ductile** (layers of ions slide but electrons keep holding them together); shiny

Nanoparticles

Nanoscience is the study of structures **1–100 nm** in size (1 nm = 1×10^{-9} m), containing a few hundred to a few thousand atoms. Compare this with atoms and small molecules (~0.1–0.5 nm, i.e. 10^{-10} m) and with fine dust / "coarse particles" (PM10, 2500–10 000 nm).

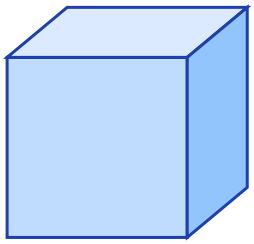
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Worked Example: A silver nanoparticle has a diameter of 1.6×10^{-8} m; a silver atom has a diameter of 2.9×10^{-10} m. Roughly how many atoms wide is the nanoparticle?
$1.6 \times 10^{-8} \div 2.9 \times 10^{-10} = 0.55 \times 10^{2} \approx$ **55 atoms across** (an order of magnitude estimate: $\sim 10^{2}$).

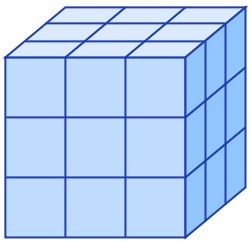
Surface Area to Volume Ratio

As particles get **smaller**, their **surface area to volume ratio increases**. For a cube of side *L*: surface area = $6L^2$, volume = L^3, so SA : V = 6/L. Dividing the side length by 10 multiplies the SA : V ratio by 10.

Smaller Particles Have a Larger Surface Area to Volume Ratio



one 3 cm cube
SA 54 cm² · V 27 cm³ · ratio 2 : 1



cut into 27 × 1 cm cubes
SA 27 × 6 = 162 cm² · V 27 cm³ · ratio 6 : 1

Same total volume,
3 × the surface area.

A 10 nm cube has an SA : V ratio 1 000 000 × bigger than a 1 cm cube.

Worked Example: Calculate the surface area to volume ratio of a cube of side 2 nm and of side 20 nm.
2 nm cube: SA = $6 \times 2^2 = 24$ nm²; V = $2^3 = 8$ nm³; ratio = 24 / 8 = **3 : 1** (or 3 nm^{-1}).
20 nm cube: SA = $6 \times 20^2 = 2400$ nm²; V = $20^3 = 8000$ nm³; ratio = 2400 / 8000 = **0.3 : 1**.
The side length is 10 times smaller, so the SA : V ratio is **10 times bigger**.

How this affects properties: a much larger fraction of the atoms in a nanoparticle are **on the surface**, where they can react. So nanoparticles:

- are much more **reactive** and make very effective **catalysts** — smaller quantities are needed than of the bulk material;
- dissolve and react faster (e.g. powders vs lumps; crushed vs whole tablets);

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- can have **different properties** from the same material in bulk (e.g. nanoparticles of gold are red / purple and catalytic; nanoparticles of titanium dioxide are transparent but still absorb UV light).

> **PAG C8 link — dissolving tablets:** compare how long whole, halved, quartered and powdered effervescent (or indigestion) tablets take to dissolve / stop fizzing in water. Smaller pieces have a larger total surface area, so more particles are exposed and the reaction is faster. Keep the volume and temperature of water the same; time with a stopwatch, or measure mass loss of CO_2 on a balance. (Rates are developed fully in Topic C5.)

Uses of Nanoparticulate Materials

Nanoparticle	Property	Use
Titanium dioxide / zinc oxide	Absorb / block UV light but are transparent (no white film on skin)	Sun creams; self-cleaning glass
Silver nanoparticles	Antibacterial; large surface area	Wound dressings, socks, food-storage containers, deodorants
Carbon nanotubes	Very strong and light; conduct electricity	Reinforcing sports equipment; electronic circuits; possible drug delivery
Fullerenes (C_{60} cages)	Hollow cage; can carry other molecules	Delivering drugs to target cells; lubricants
Metal nanoparticles (Pt, Pd, Au)	Huge surface area → highly catalytic	Catalysts (e.g. catalytic converters, fuel cells)
Nanoparticles in cosmetics	Small, smooth, absorbed easily	Moisturisers, deodorants

Possible Risks of Nanoparticles

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- Their small size means they could be **breathed in** or **absorbed through the skin** and pass into cells and the bloodstream, where they might cause damage (e.g. lung inflammation).
- Their high surface area / reactivity could make them **toxic** or catalyse harmful reactions in the body.
- They can be **released into the environment** (e.g. silver from washed clothes, sun cream into the sea) and may harm aquatic organisms or useful bacteria.
- Their **long-term effects are not yet known**, because they have only been used widely for a short time — scientists disagree, so more research and regulation (e.g. labelling) is needed.

> **Exam Tip:** For "evaluate the use of nanoparticles" questions, give **both sides**: the benefit (linked to a specific property, e.g. large SA:V → better catalyst; transparent UV-blocker in sun cream) **and** the risk (unknown long-term effects; can enter cells / the environment), then reach a justified conclusion.

Quick Summary — Key Ideas for C2

Idea	Key point / equation
Purity	Pure = one element or compound; sharp m.p. Impure: melts over a range, below the pure m.p.
Relative formula mass	M_r = sum of A_r values; total M_r of reactants = products
Empirical formula	Simplest whole-number ratio of atoms
Separation	Filtration (insoluble solid) · crystallisation (soluble solid) · simple distillation (solvent) · fractional distillation (miscible liquids) · chromatography (dissolved mixtures)
Chromatography	R_f = distance moved by substance ÷ distance moved by solvent; pure = one spot
Periodic Table	Group = outer electrons; period = number of shells; order by atomic number

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Bonding	Ionic: transfer (metal + non-metal). Covalent: sharing (non-metals). Metallic: ions + delocalised electrons
Melting points	Simple molecules low (weak intermolecular forces); ionic, giant covalent and metals high
Carbon	4 bonds; chains and rings; diamond (4 bonds, hard), graphite (3 bonds, layers, conducts), graphene, fullerenes
Nanoparticles	1–100 nm; SA : V = 6/L for a cube; large SA : V → new properties and uses; uncertain risks

Specification Checklist

Code	You should be able to...	Tier
C2.1a	Explain purity; distinguish scientific and everyday use of "pure"	F & H
C2.1b	Use melting point data to distinguish pure from impure substances	F & H
C2.1c	Calculate relative formula masses, separately and in a balanced equation; define A_r, relative molecular and relative formula mass	F & H
C2.1d	Deduce the empirical formula from numbers of atoms or a model / diagram, and vice versa	F & H
C2.1e	Explain that many useful materials are formulations of mixtures (e.g. alloys)	F & H
C2.1f	Describe, explain and exemplify filtration, crystallisation, simple and fractional distillation	F & H
C2.1g	Describe paper and thin layer chromatography (aqueous / non-aqueous solvents, locating agents)	F & H
C2.1h	Recall stationary and mobile phases; separation depends on distribution between phases	F & H

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C2.1i	Interpret chromatograms, including measuring R_f values	F & H
C2.1j	Suggest suitable purification techniques	F & H
C2.1k	Suggest chromatographic methods (paper, TLC, GC) to distinguish pure from impure substances	F & H
C2.2a	Describe metals and non-metals and explain their differences (physical properties, ions, oxides)	F & H
C2.2b	Explain how the atomic structure of metals and non-metals relates to their position in the Periodic Table	F & H
C2.2c	Relate position in the Periodic Table to electron arrangement and atomic number (group and period)	F & H
C2.2d	Describe and compare bonds in ionic compounds, simple molecules, giant covalent structures, polymers and metals	F & H
C2.2e	Explain bonding in terms of electrostatic forces and transfer / sharing of electrons	F & H
C2.2f	Construct dot and cross diagrams for simple covalent and binary ionic substances	F & H
C2.2g	Describe limitations of dot and cross, ball and stick, 2D and 3D representations	F & H
C2.2h	Explain how reactions of elements relate to electron arrangement and atomic number	F & H
C2.2i	Explain how Mendeleev's arrangement was refined into the modern Periodic Table using atomic number	F & H
C2.3a	Recall that carbon can form four covalent bonds	F & H
C2.3b	Explain the vast array of organic compounds (families, chains and rings)	F & H

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C2.3c	Explain properties of diamond, graphite, fullerenes and graphene from structure and bonding	F & H
C2.3d	Use energy transfers and strength of bonds / intermolecular forces to explain melting and boiling temperatures	F & H
C2.3e	Use data to predict states of substances under given conditions	F & H
C2.3f	Explain bulk properties of the five structure types; atoms themselves don't have bulk properties	F & H
C2.3g	Compare "nano" dimensions with atoms and molecules	F & H (separate chemistry)
C2.3h	Describe the surface area to volume relationship and how it affects properties	F & H (separate chemistry)
C2.3i	Describe how properties of nanoparticulate materials relate to their uses	F & H (separate chemistry)
C2.3j	Explain possible risks of some nanoparticulate materials	F & H (separate chemistry)

Practical Activity Groups Linked to C2

PAG	Title	Link to C2
PAG C3	Separation techniques	Paper / thin layer chromatography of an unknown ink (C2.1g, C2.1k)
PAG C4	Distillation	Simple / fractional distillation of mixtures; purification (C2.1a, C2.1f)
PAG C7	Production of salts	Purifying compounds by filtration and crystallisation (C2.1a, C2.1f)

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PAG C8	Measuring rates of reaction	Dissolving tablets — effect of surface area (C2.3h)

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